5 **(a)** current (through a conductor is directly) proportional to potential difference (across the conductor) or vice versa
(provided that) temperature (of conductor remains) constant [1]

(b) **(i)** $R = \rho L / A$
$\rho = (18 \times 7.2 \times 10^{-8}) / 0.94 = 1.4 \times 10^{-6}\ \Omega$ m [1]

(ii) current in the battery: increase [1]
voltmeter reading: decrease [1]

(iii) cross marked on the resistance wire to right of the arrowhead of S, but not touching the right-hand end of the resistance wire [1]

(c) **(i)** $I = Anvq$
$q = 0.93 / [(7.2 \times 10^{-8}) \times (9.0 \times 10^{28}) \times (1.3 \times 10^{-3})]$
$q = 1.1 \times 10^{-19}$ C [1]

(ii) charge / q (value) is below 1.6×10^{-19} (C) [1]
or
charge cannot be below 1.6×10^{-19} (C)
or
(the charge carriers / q) should have a charge of 1.6×10^{-19} (C)

(d) Before t_1 / when current constant, the (total) resistance is constant

At t_1 / when current increases, the (total) resistance decreases (due to decrease of external resistance)

After t_1, temperature (of lamp) increases so the resistance of the lamp increases (so total resistance increases so the current in the ammeter decreases)
[3]